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Bank Loan Dashboard

Overview

The Comprehensive Bank Loan Dashboard is an innovative project designed to enhance the analysis and reporting of loan application data. By utilizing Microsoft Excel for data preparation and Power BI for visualization, this dashboard provides stakeholders with valuable insights, facilitating informed decision-making across teams.

Tools Used

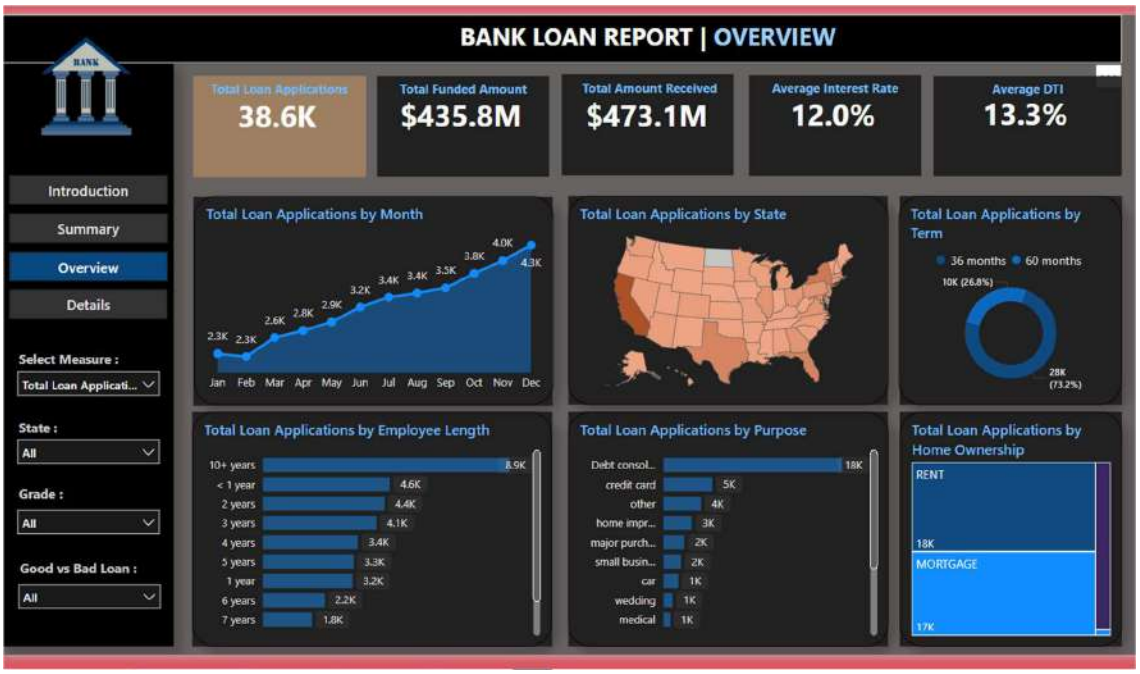
- **Microsoft Excel**:
  - Initial data cleaning and organization.
  - Preliminary analysis of loan application data.
- **Power BI**:
  - Creation of dynamic visualizations and an interactive dashboard.
  - Transformation of complex datasets into easily interpretable visuals.

Key Features

1. **Total Loan Applications by Month**:
   - A line chart tracking monthly loan applications.
   - Helps identify seasonal trends in loan demand.
2. **Geographic Breakdown by State**:
   - Map visualization displaying loan applications by state.
   - Supports targeted marketing strategies based on regional performance.
3. **Loan Terms Analysis**:
   - Bar chart illustrating the distribution of loan terms (15, 20, 30 years).
   - Enhances understanding of customer preferences regarding loan durations.

4. **Purpose of Loans**:
    - Stacked column chart categorizing applications by intended use (home purchase, refinancing, personal loans).
    - Aligns offerings with customer needs.
5. **Home Ownership Status**:
    - Donut chart visualizing the correlation between home ownership status (owned, mortgaged, rented) and loan approval rates.
    - Provides insights into borrower demographics.
6. **Good vs. Bad Loans**:
    - Pie charts showing the ratio of successful loans to high-risk loans.
    - Contributes to ongoing risk assessment initiatives.
7. **Loan Status Overview**:
    - Card visualization summarizing the status of all applications (pending, approved, rejected).
    - Enables quick access to critical information.

Loan Status	Total Loan Applications	Total Funded Amount	Total Amount Received	MTD Funded Amount	MTD Received Amount	Avg. Interest Rate	Avg. DTI
Fully Paid	32145	$35,13,58,350	$41,15,86,256	$4,13,02,025	$4,78,15,851	11.64%	13.17%
Charged Off	5333	$6,55,32,225	$3,72,84,763	$87,32,775	$53,24,211	13.88%	14.00%
Current	1098	$1,88,66,500	$2,41,99,914	$39,46,625	$49,34,318	15.10%	14.72%
Total	38576	$43,57,57,075	$47,30,70,933	$5,39,81,425	$5,80,74,380	12.05%	13.33%



ID	Purpose	Home Ownership	Grade	Sub Grade	Issued Date	Funded Amount	Interest Rate	Installment	Received Amount
632417	small business	OWN	F	F1	11 May 2021	$35,000	0.20	932.17	$10,430
653921	renewable_energy	MORTGAGE	D	D5	11 March 2021	$35,000	0.16	851.51	$39,846
668166	credit card	MORTGAGE	G	G2	11 February 2021	$35,000	0.20	936.66	$53,494
670409	Debt consolidation	RENT	F	F3	11 February 2021	$35,000	0.19	907.73	$49,122
672549	credit card	MORTGAGE	E	E3	11 February 2021	$35,000	0.17	872.48	$52,349
672680	Debt consolidation	OWN	E	E1	11 February 2021	$35,000	0.16	858.59	$35,479
673070	Debt consolidation	MORTGAGE	E	E4	11 February 2021	$35,000	0.18	879.47	$3,511
673667	credit card	MORTGAGE	D	D5	11 February 2021	$35,000	0.16	851.51	$41,509
674426	Debt consolidation	MORTGAGE	F	F1	11 March 2021	$35,000	0.18	893.54	$53,612
674448	Debt consolidation	MORTGAGE	G	G2	11 February 2021	$35,000	0.20	936.66	$56,199
674846	home improvement	MORTGAGE	C	C4	11 February 2021	$35,000	0.14	1,192.83	$41,819
674982	Debt consolidation	MORTGAGE	G	G4	11 February 2021	$35,000	0.21	951.21	$14,374
675763	home improvement	MORTGAGE	E	E4	11 February 2021	$35,000	0.18	879.47	$43,703
676268	Debt consolidation	MORTGAGE	C	C1	11 February 2021	$35,000	0.13	1,173.91	$41,341
676756	Debt consolidation	MORTGAGE	E	E3	11 February 2021	$35,000	0.17	872.48	$49,760
678088	home improvement	MORTGAGE	E	E4	11 March 2021	$35,000	0.18	879.47	$41,630
678292	Debt consolidation	MORTGAGE	E	E2	11 February 2021	$35,000	0.17	865.52	$45,468

Benefits

- **Real-time Analysis**: Stakeholders can access up-to-date information, promoting quicker decision-making.
- **Data-Driven Decisions**: Visualizations highlight key performance metrics, aiding in strategic planning and resource allocation.
- **User-Friendly Interface**: The interactive nature of the dashboard allows team members to drill down into specific data points for deeper analysis.

Conclusion

The Comprehensive Bank Loan Dashboard has significantly enhanced reporting capabilities and provides a clear overview of the loan application landscape. The project has fostered collaboration among team members, and their support was invaluable throughout the development process. This dashboard not only streamlines data analysis but also empowers the organization to make informed, strategic decisions based on real-time insights.